Robotic Manipulation

Perception, Planning, and Control

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Last modified 2025-8-26.
How to cite these notes, use annotations, and give feedback.

Note: These are working notes used for a course being taught at MIT. They will be updated throughout the Fall 2024 semester.

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Appendix E
Miscellaneous

E.1 How to cite these notes

Thank you for citing these notes in your work. Please use the following citation:

```
@book{manipulation,
  title        = "Robotic Manipulation",
  subtitle     = "Perception, Planning, and Control",
  howpublished = "Course Notes for MIT 6.421",
  author       = "Tedrake, Russ",
  year         = 2024,
  url          = "http://manipulation.mit.edu",
}
```

E.2 Annotation tool etiquette

My primary goal for the annotation tool is to host a completely open dialogue on the intellectual content of the text. However, it has turned out to serve an additional purpose: it's a convenient way to point out my miscellaneous typos and grammatical blips. The only problem is that if you highlight a typo, and I fix it 10 minutes later, your highlight will persist forevermore. Ultimately this pollutes the annotation content.

There are two possible solutions:

- you can make a public editorial comment, but must promise to delete that comment once it has been addressed.
- You can join my "editorial" group and post your editorial comments using this group "scope".

Ideally, once I mark a comment as "done", I would appreciate it if you can delete that comment.

I highly value both the discussions and the corrections. Please keep them coming, and thank you!

E.3 Some great final projects

Each semester students put together a final project using the tools from this class. So many of them are fantastic! Here is a small sample that captures some of the diversity.

Fall 2023 Outstanding Project Awards:

- OrderBot: Food Preparation Manipulation Bot for Fast Food Environments by Shreya Chaudhary,
- APPLE-E: A Compliant Apple Picking Robot by Udayan Bulchandani,
- Guitar Hero Bot: Motion Planning to Solve Rhythm Games with Robotic Manipulation by Ryan Hourican, John Readlinger, and Noah Fisher,
- An Autonomous Pool-Playing Robot by Jinger Chong and Eugenia Feng,
- Dreidel Spinning Bot: Exploring Contact Models in Simulation by Lily Yeazell,
- The Robot Librarian: Mobile Manipulation in a Library by Shiva Sreeram and Peter Holderrieth,
- CatchingBot: A Robotic System for Catching Objects in Flight (including online estimation/replanning) by Bart艂omiej Cieslar and Michael Zeng,
- ShelveBot: Fast Pick-and-Place in Human Form Factors by Julianna Schneider and Shayan Pardis and Anna Yang.

And here is a playlist of all the 2023 projects which opted in to being public.

Fall 2022 Outstanding Project Awards (playlist):

- ChessBot by Ethan Chung,
- Rock Skipping Robot by Michael Burgess and Nicholas Ramirez,
- How we made a robot arm juggle! by Richard Li, David Jin, and Shao Yuan Chew Chia,
- SpeedCuber Bot by Bin Pham,
- Michaelangelo - Robot Painter 2.0 by Vainavi Mukkamala and Akila Saravanan,
- Ping Pong Bot 2023 by Jenny Zhang and Daniel Klahn,
- TetrisBot by Pranav Arunandhi, Ashley Ke, and Sadhana Lolla,
- Multiple-finger Grasping: Generating Antipodal Grasping with Allegro Hand by Yuxiang Ma,
- Exploring 6DOF Pose Estimation Approaches Using RGB(D) by Amin Heyrani Nobari.

Fall 2021:

- YouTube playlist containing all projects that opted in.
- Playing Piano with a Robotic Hand by Seong Ho Yeon
- Table Cleaning through Task and Motion Planning with Force Control by William Shen
- RallyBot: Exploring Physics-Based Approaches to Robotic Table Tennis by Dylan Zhou and Chaitanya Ravuri
- Robust Stacking of Convex Polytopes by Richard Li and Gabriel Margolis
- Simulation of Discrete Lattice Robot Assemblies by Miani Smith
- Da Vinci Robot by Maisy Lam and Jose A. Muguira
- Certifying Convex Decompositions of Free Configuration Space by Alexandre Amice, Peter Werner, Annan Zhang
- Robotic Arm Weightlifting via Trajectory Optimization by Timur Garipov
- An Efficient Implementation of Pressure Field Models by Vincent Huang, Franklyn Wang, Eric Zhang
- Joint Perception and Manipulation System for Multi-Shape Peg-in-Hole Tasks by Portia Gaitskell and Julia Wyatt

Fall 2020:

- Solving Simple Tiling Puzzles with an End-to-End Robotic System by Anubhav Guha
- Throwing in Drake by Daniel Yang and Tony Wang
- Robot Juggler by Ryan Shubert and Matt Beveridge
- A Geometric Approach to Recycling Cans via Vision-Based Manipulation by Susan Ni and George Chen
- Interactive Perception: Scene Segmentation Through Physical Perturbation by Lukas Lao Beyer and John Keszler
- William Shakebot by Carson Smithbot and Clemente Ocejo

E.4 Please give me feedback!

I'm very interested in your feedback. The annotation tool is one mechanism, but you can also comment directly on the YouTube lectures, or even add issues to the github repo that hosts these course notes.

I will also post a proper survey questionnaire here once the notes/course has existed long enough for that to make sense.
